chapter: 8 Simple Linear Regression and Correlation

Pre-Board, NAST: 2022-Fall / 2020-Fall

① The following table gives the ages and blood pressure of 10 women:

Age(X):	56	42	36	47	49	42	60	72	63	55
BP(Y):	147	125	118	128	145	140	155	160	149	150

i) Find the correlation coefficient between age and pressure.
ii) Determine the least square regression of Y on X.
iii) Estimate the BP of a women whose age is 45 yrs.

sol Here,

$n = 10$

X	Y	X^2	XY	Y^2
56	147	3136	8232	21609
42	125	1764	5250	15625
36	118	1296	4248	13924
47	128	2209	6016	16384
49	145	2401	7105	21025
42	140	1764	5880	19600
60	155	3600	9300	24025
72	160	5184	11520	25600
63	149	3969	9387	22201
55	150	3025	8250	22500
522	1417	28348	75188	202493

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i> Correlation coefficient between age and pressure:

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$= \frac{10 \times 75188 - 522 \times 1417}{\sqrt{10 \times 28348 - (522)^2}\sqrt{10 \times 202493 - (1417)^2}}$$

$\therefore r = 0.8917$ Ans

ii> Here,

Let $Y = a + bX$ ——— (1)

be the least square regression of Y on X.

The values of a and b are obtained by solving these equations,

$$\sum Y = na + b\sum X \quad \text{——— (2)}$$

$$\sum XY = a\sum X + b\sum X^2 \quad \text{——— (3)}$$

Now,

$$1417 = 10a + 522b$$

$$75188 = 522a + 28348b$$

On solving these equestions, we get;-

$\therefore a = 83.76;\ 1.11$

From (1),

$\therefore Y = 83.76 + 1.11X$

This is the required least square regression of Y on X. Ans

iii) NRO,

$$Y = 83.76 + 1.11 x$$

For age = 45,

$$\therefore Y = 83.76 + 1.11 \times 45 \approx 133.71 \approx 134$$

Hence, the estimated blood pressure of women whose age is 45 yrs ≈ 134 Ans.

2021-Spring

23. In the US, during the fall harvest season Pumpkins are sold in the large quantities at the form stands often instead of weighting the Pumpkin prior to scale, the farm stand operator will just place the Pumpkin in the appropriate circular cut out on the counter. We asked why this was done, one farmer replied, "I ~~am~~ can tell the weight of the pumpkin from its circumference". To determine whether this was reality true; a sample of 8 pumpkin's circumference and weights were measured.

Circumference in cm:	50	55	54	52	37	52	53	47
weight in grams(00):	12	20	15	17	9	14	16	14

i> Develop the estimating linear regression equation to predict the weight of the pumpkin that is 60cm in circumference.

ii> Calculate the standard error of the estimate and interpret the result.

iii> What percentage of the total variance in weight of the pumpkin is explained by the circumference of the pumpkin?

Sol

Let x = circumference in cm

y = weight in grams (00)

i) Here,

let $Y = a + bx$ ——— (1)

be the linear regression equation.

The values of a and b are obtained by solving these equations,

$$\Sigma Y = na + b\Sigma X \quad \text{——— (2)}$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \quad \text{——— (3)}$$

where,

$n = 8$

Now,

X	Y	XY	X^2	Y^2
50	12	600	2500	144
55	20	1100	3025	400
54	15	810	2916	225
52	17	~~714~~ 884	2704	289
37	9	333	1369	81
52	14	728	2704	196
53	16	848	2809	256
47	14	658	2209	196
400	117	5961	~~19296~~ 20236	1787

from (2) & (3),

~~117 = 8a + 400b~~ $117 = 8a + 400b$

~~5961 = 340 + 19296~~ $5961 = 400a + 20236b$

on solving these equations,

$a \simeq -8.89$; $b \simeq 0.47$

From (i),

$$Y = -8.89 + 0.47X$$

This is the required linear regression equation.

Now,

for circumference of pumpkin = 6 cm.

$$\therefore Y = -8.89 + 0.47 \times 60 = \cancel{6.07(00)}\ gm$$

$$= \cancel{607\ gm}\ 19.31\ gm \quad \text{Ans}$$

Hence, the weight of pumpkin whose weight is 6cm ≈ -607 gm. Ans

ii) Standard Error, $SEE = \sqrt{\dfrac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$

$$= \sqrt{\dfrac{1787 - (-8.89) \times 117 - 0.47 \times \cancel{4000}5961}{8-2}}$$

$$= \cancel{2.05}\ 2.06 \quad \text{Ans}$$

Interpretation:

The ~~avera~~ standard error 2.06 tells us that average data falls 2.06 units from the regression line.

iii> To find percentage of variance, we have to calculate coefficient of determination.

$$r^2 = \left(\frac{n\sum XY - \sum X \sum Y}{\sqrt{n\sum X^2 - (\sum X)^2} \cdot \sqrt{n\sum Y^2 - (\sum Y)^2}} \right)^2$$

$$= \left(\frac{8 \times 5961 - 400 \times 117}{\sqrt{8 \times 20236 - (400)^2} \cdot \sqrt{8 \times 1787 - (117)^2}} \right)^2$$

$$\therefore r^2 = 0.688 \approx 68.80\ \%$$

Hence the percentage of the variance is 68.80%

Ans

2021 - Fall

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6. b) A chemical company wishing to study the effect of extraction time on the efficiency of an extraction operation obtained as follows:

Extraction time in minute (x): 27, 45, 41, 19, 35, 39, 19

Extraction efficiency in % (y): 57, 64, 80, 46, 62, 72, 52

i) Find the regression equation of y on x and estimate the y if x = 40.

ii) Find the coefficient of correlation between x and y

iii) Determine coefficient of determination and interpret

iv) Compute standard error of estimate.

Sol Here,

$n = 7$

x	y	xy	x^2	y^2
27	57	1539	729	3249
45	64	2880	2025	4096
41	80	3280	1681	6400
19	46	874	361	2116
35	62	2170	1225	3844
39	72	2808	1521	5184
19	52	988	361	2704
225	433	14539	7903	27593

i) Here,

Let $y = a + bx$ ——— (1)

be the regression equation of y on x.

0.8432

The values of a and b are obtained by solving these equations,

$\Sigma y = na + b\Sigma x$ ——— (2)

$\Sigma xy = a\Sigma x + b\Sigma x^2$ ——— (3)

From (2) & (3),

$433 = 7a + 225b$

$14539 = 225a + 7903b$

on solving these equations, we get:-

$a = 32.09$; $b = 0.93$

From (1),

$y = 32.09 + 0.93x$

For $x = 40$,

$\therefore y = 32.09 + 0.93 \times 40 = 69.29$ Ans

ii) coefficient of correlation between x and y :

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{7 \times 14539 - 225 \times 433}{\sqrt{7 \times 7903 - (225)^2}\sqrt{7 \times 27593 - (433)^2}}$$

$\therefore r = 0.8432$ Ans

iii) Coefficient of determination $= r^2$

$$= (0.8432)^2$$

$$= 0.7110$$

Interpretation: The coefficient of determination 0.7110 tells us that the percentage of the total variance of the extraction efficiency in % is 71.10 %.

iv) Sandard error of estimate,

$$SEE = \sqrt{\frac{\sum y^2 - a\sum Y - b\sum XY}{n-2}}$$

$$= \sqrt{\frac{27593 - 32.09 \times 433 - 0.93 \times 14539}{7-2}}$$

$\therefore SEE = 5.946$ Ans

2022-Fall

Q.6.b) The following table presents shear strength (in KN/mm) and weld diameters (in mm) for a sample of spot welds.

Diameter	4.2	4.4	4.6	4.8	5.0	5.2	5.4
Strength	51	54	69	81	75	79	89

i) Compute the least-squares line for predicting strength from diameter.

ii) Predict the strength for a diameter of 5.5 mm.

Sol Here,

$n=7$; Let y = strength ; x = diameter

x	y	xy	x^2	y^2
4.2	51	214.2	17.64	2601
4.4	54	237.6	19.36	2916
4.6	69	317.4	21.16	4761
4.8	81	388.8	23.04	6561
5.0	75	375	25	5625
5.2	79	410.8	27.04	6241
5.4	89	480.6	31.36	7921
33.6	498	2424.4	162.4	36626

i) Let $y = a + bx$ ——— (1)

be the least squares line for predicting strength from diameter.

The values of a and b are obtained by solving these equations,

$$\Sigma y = na + b\Sigma x \quad \text{---} (2)$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 \quad \text{---} (3)$$

From (2) & (3), we get;

$$498 = 7a + 33.6b$$
$$2424.4 = 33.6a + 162.4b$$

On solving these equations, we get;-

$$a = -74.57$$
$$b = 30.36$$

From (1),

$$y = -74.57 + 30.36x$$

This is the required least square-regression line for predicting strength from ~~diag~~ diameter.

Ans.

ii> we have, diameter = 5.5 mm.

$$\therefore y = -74.57 + 30.36 \times 5.5 = 92.41 \text{ KN/mm}$$

Hence, the required strength is 92.41 KN/mm. Ans

2023-Spring

6.b) The following are marks obtained by 9 students in Mathematics (X) and statistics (Y):

X:	40	65	60	25	85	35	45	70	80
Y:	30	85	65	35	90	85	75	75	45

i) Develop the estimating regression equation of X on Y.
ii) Find the correlation coefficient
iii) Find the coefficient of determination.

Here,

X	Y	XY	X^2	Y^2
40	30	1200	1600	900
65	85	5525	4225	7225
60	65	3900	3600	4225
25	35	875	625	1225
85	90	7650	7225	8100
35	35	1225	1225	1225
45	75	3375	2025	5625
70	75	5250	4900	5625
80	45	3600	6400	2025
505	535	32600	31825	36175

where, $n = 9$

i) Let $X = a + bY$ ——— (1)
be the regression equation of X on Y.

The values of a and b are obtained by solving these equations,

$$\Sigma X = na + b\Sigma Y \quad \text{———} (2)$$
$$\Sigma XY = a\Sigma Y + b\Sigma Y^2 \quad \text{———} (3)$$

from (2) & (3),

$$505 = 9a + 535b$$
$$32600 = 535a + 36175b$$

On solving these equations, we get:-

$$a = 21.03 \; ; \; b = 0.59$$

from (1),

$$X = 21.03 + 0.59Y$$

This is the required regression of X on Y. Ans

ii> The coerralation coefficien:

$$r = \frac{n\Sigma XY - \Sigma X \Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2} \cdot \sqrt{n\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{9 \times 32600 - 505 \times 535}{\sqrt{9 \times 31825 - (505)^2} \sqrt{9 \times 36175 - (535)^2}}$$

$$\therefore r = 0.66$$ Ans

iii) coefficient of determination $= r^2$

$= (0.66)^2$

$= 0.4356$ Ans

2018 - Spring

20. A film administers a test to sales trainees before they go into the field. The management of the firm is interested in determining the relationship between the test scores and the sales made by the trainees at the end of one year in the field. The following data were collected for 14 sales personnel who have been in the field one year.

salesperson:	1	2	3	4	5	6	7	8	9	10	11	12	13	14
TS (X):	26	37	24	45	26	50	28	30	40	34	29	30	43	38
No. of units (Y):	9	14	8	18	10	19	11	13	17	15	13	16	20	17

calculation shows that $\Sigma X = 480$, $\Sigma X^2 = 17296$.

$\Sigma Y = 200$, $\Sigma Y^2 = 3044$

$\Sigma XY = 7215$

i> Find the correlation coefficient between the test ~~and~~ scores and number of units sold.

ii> Assuming a linear relationship, use the least squares method to find the regression equation and interpret the meaning of y-intercept and slop.

iii> Compute the residual for salesperson 9.

iv> Compute coefficient of determination.

Sol^n Here,

$\Sigma X = 480$; $\Sigma X^2 = 17296$

$\Sigma Y = 200$; $\Sigma Y^2 = 3044$

$\Sigma XY = 1715$

i) The correlation coefficient between the test scores and number of units sold:

$$r = \frac{n\Sigma XY - \Sigma X \Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2}\sqrt{n\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{14 \times 7215 - 480 \times 200}{\sqrt{14\times 17296 - (480)^2}\sqrt{14 \times 3044 - (200)^2}}$$

$\therefore r = 0.90388$ Ans

ii) Let $Y = a + bX$ ——— (1)
be the least square equation.

The values of a and b are obtained by solving these equations,

$$\Sigma Y = na + b\Sigma X \quad \text{——— (2)}$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \quad \text{——— (3)}$$

From (2) & (3), we get;

$$200 = 14a + 480b$$

$$7215 = 480a + 17296b$$

On solving these equations, we get:-

~~a = 224.455; b = -6.130~~

$$\boxed{a = -0.34 \; ; \; b = 0.43}$$

~~From (1), Y = 224.455 - 6.130X~~

This i

From (1),

$$Y = -0.34 + 0.43x$$

This is the required regression line. Ans

Interpretation:

'a' is called y-intercept, which represent the value of y when $x = 0$.

'b' is the slope of regression line y on x.

iii> The residual for salesperson 9 is:

$$\text{Residual} = y - \hat{y}$$
$$= 17 - (-0.34 + 0.43 \times 40)$$
$$= 0.14$$

$\therefore$ Residual $= 0.14$ Ans

iv> Coefficient of determination is the square of coefficient of correlation i.e. r^2

$$= (0.90388)^2$$
$$= 0.816999$$

Ans

2019-Spring

18) An engineering student has a summer job with the forestry service. He measured the tree trunk diameters (x in inches) and related them to the age of the tree (y in years). The following information was obtained. $n=6$, $\Sigma x=21$, $\Sigma Y=26$, $\Sigma x^2=91$, $\Sigma Y^2=142.52$, $\Sigma XY=113.8$.

i) Find the regression equation of y on x and estimate the age of the tree whose diameter is 4 inches.

ii) Find the correlation coefficient between x and y and interpret its meaning.

iii) Determine coefficient of determination and interpret it.

iv) Compute standard error of the estimate.

Sol Given;

$n=6$

$\Sigma X=21$; $\Sigma X^2=91$

$\Sigma Y=26$; $\Sigma Y^2=142.52$

$\Sigma XY=113.8$

i) Let $Y=a+bX$ ——— (1)

be the regression equation of Y on X.

The values of a and b are obtained by solving these two equations,

$$\Sigma Y = na + b\Sigma X \quad \text{——— (2)}$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \quad \text{——— (3)}$$

From (2) & (3), we get;-

$26 = 6a + 21b$

$113.8 = 21a + 91b$

on solving these equations, we get;-

$a = -0.227$; $b = 1.303$

From (1),

$Y = -0.227 + 1.303x$

This is the required regression equation of y on x. Ans

For $x = 4$ inches,

$y = -0.227 + 1.303 \times 4$

$\therefore y = 4.985 \simeq 5$ years.

Hence, the age of tree whose diameter is 4 inches $\simeq$ 5 years. Ans

ii) The coefficient of correlation between x and y is,

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{6 \times 113.8 - 21 \times 26}{\sqrt{6 \times 91 - (21)^2}\sqrt{6 \times 142.52 - (26)^2}}$$

$\therefore r = 0.9975$ Ans

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iii) coefficient of determination is equal to the square of correlation coefficient i.e. r^2

$$= 0.9975^2$$

$$= 0.995 \quad \text{Ans}$$

iv) standard error of estimate, $SEE = \sqrt{\dfrac{\Sigma y^2 - a\Sigma y - b\Sigma xy}{n-2}}$

$$= \sqrt{\dfrac{142.52 - (-0.227)\times 26 - 1.303 \times 113.8}{6-2}}$$

$$\therefore SEE = 0.187 \quad \text{Ans}$$